Assemble and Disassemble a PC Practical

1.

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

1. **Motherboard:** The motherboard is the main circuit board of the computer. It connects all hardware components, such as the CPU, RAM, storage drives, and GPU.
2. **CPU (Central Processing Unit):** The CPU is the brain of the computer. It processes instructions, performs calculations, and controls many of the computer's operations.
3. **RAM (Random Access Memory):** RAM temporarily stores the data and programs currently being used by the computer. More RAM helps the computer handle multiple applications smoothly.
4. **GPU (Graphics Processing Unit):** The GPU processes images, videos, and graphics. It is especially important for gaming, video editing, and graphic design.
5. **HDD (Hard Disk Drive):** An HDD is a storage device that uses spinning magnetic disks to store the operating system, software, and personal files.
6. **SSD (Solid-State Drive):** An SSD stores data using flash memory and has no moving parts. It provides faster boot times and quicker application loading than a traditional HDD.
7. **Power Supply Unit (PSU):** The PSU converts electricity from the wall outlet into the appropriate power required by the computer's components.
8. **CPU Cooling System (Fan and Heatsink):** The CPU fan and heatsink remove heat from the processor. They help prevent overheating and maintain stable performance.
9. **Cabinet Fans:** Cabinet fans circulate air inside the computer case. They help remove hot air and keep internal components cool.
10. **Optical Drive (CD/DVD Drive):** An optical drive reads or writes CDs and DVDs. It is optional and is not included in many modern desktop PCs.
11. **Expansion Cards:** Expansion cards are installed in motherboard expansion slots to add features such as improved graphics, Wi-Fi, sound, or additional ports.
12. **CMOS Battery:** The CMOS battery supplies power to the motherboard's real-time clock and helps retain BIOS/UEFI settings when the computer is switched off.
13. **SATA Data Cables:** SATA cables connect compatible storage devices, such as HDDs and SATA SSDs, to the motherboard for data transfer.
14. **Power Cables:** Power cables connect the PSU to the motherboard, CPU, GPU, and storage devices to supply the required electricity.
15. **PC Cabinet (Computer Case):** The cabinet houses and protects the internal hardware components. It also provides mounting points for components and supports airflow.

2.

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



3.

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Steps to Reassemble the PC

Step 1: Prepare the PC Case

Place the computer case on a clean, flat surface. Ensure that the power cable is unplugged and touch an unpainted metal part of the case to reduce static electricity.

Step 2: Install the Motherboard

Carefully place the motherboard inside the case and secure it with screws. Ensure that the motherboard is correctly aligned with the mounting holes to prevent damage.

Step 3: Install the CPU and Cooling System

Check that the CPU is correctly seated in its socket. Install the CPU cooler and connect its fan cable to the CPU_FAN header on the motherboard.

Step 4: Install RAM
Insert the RAM modules into the appropriate memory slots. Press gently until the clips lock into place. Installing RAM before closing the case makes it easier to check that the modules are properly seated.

Step 5: Install the Storage Drive
Reinstall the HDD or SSD in its designated location. Connect the SATA data and power cables, or secure an M.2 SSD in its slot.

Step 6: Install the Power Supply and Connect Power Cables
Install the power supply unit (PSU), if it was removed. Connect the 24-pin motherboard power cable and the 4/8-pin CPU power cable. These connections supply power to the motherboard and processor.

Step 7: Install the Graphics Card
Insert the dedicated graphics card into the appropriate PCIe slot and secure it with screws. Connect any required PCIe power cables.

Step 8: Reconnect Other Cables
Reconnect the case's front-panel connectors, USB cables, audio cables, and case fan cables to their appropriate motherboard headers.

Step 9: Check All Connections
Verify that the RAM, storage drive, graphics card, and power cables are correctly installed. Ensure that no loose screws or cables interfere with the fans.

Step 10: Close the Case and Test the PC
Reattach the side panel and connect the monitor, keyboard, mouse, and power cable. Turn on the PC and check whether it boots correctly and detects the installed hardware.

Why Does the Order Matter?

The order of reassembly is important because some components must be installed before others become difficult to access. For example, installing the motherboard first provides a base for connecting other components, while installing RAM and the CPU cooler before closing the case makes installation easier. Connecting all power cables and checking the components before turning on the PC helps prevent connection errors and potential hardware damage.

4.
Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

A. Tools Required

S. No.	Tool	Why It Is Important
1	Phillips-head screwdriver	Used to remove and tighten screws on the PC case and hardware components.
2	Anti-static wrist strap	Helps prevent electrostatic discharge (ESD), which can damage sensitive components.
3	Anti-static mat	Provides a safer work surface for handling computer components.
4	Small container for screws	Keeps screws organized and prevents them from getting lost.
5	Cable ties	Help organize cables and prevent them from blocking cooling fans.
6	Compressed air or an electronics-safe air blower	Removes dust from fans, heatsinks, and other components.
7	Flashlight	Helps locate screws, connectors, and components inside the case.
8	Thermal paste	Used when reinstalling a CPU cooler, if the existing paste must be replaced.

B. Safety Precautions Checklist

S. No.	Safety Precaution	Why It Is Important
1	☐ Shut down the computer completely.	Prevents damage caused by working on powered components.
2	☐ Unplug the power cable.	Disconnects the PC from the main power supply and reduces electrical hazards.
3	☐ Disconnect external devices.	Prevents damage to connected devices and makes the PC easier to handle.
4	☐ Use an anti-static wrist strap or suitable ESD precautions.	Reduces the risk of electrostatic discharge damaging sensitive electronic components.
5	☐ Work on a clean, dry, and stable surface.	Prevents spills, accidental falls, and contamination from dust.
6	☐ Keep liquids and food away from the work area.	Prevents spills that could cause short circuits or damage components.

S. No.	Safety Precaution	Why It Is Important
7	☐ Handle components by their edges.	Reduces the risk of damaging delicate pins, contacts, and electronic circuits.
8	☐ Never force a component into a slot.	Prevents damage to connectors, RAM slots, CPU sockets, and other hardware.
9	☐ Keep screws and small parts organized.	Prevents lost parts and loose screws from causing short circuits inside the case.
10	☐ Handle the CPU and graphics card carefully.	Protects delicate contacts, pins, and circuit boards from physical damage.
11	☐ Never open the power supply unit (PSU).	Internal capacitors can retain dangerous electrical charges even after the PSU is unplugged.
12	☐ Double-check all connections before switching on the PC.	Helps prevent incorrect connections and potential hardware damage.

C. Final Checklist Before Starting

Before beginning PC assembly or disassembly, confirm the following:

- ☐ The computer is completely shut down.
- ☐ The power cable is unplugged.
- ☐ All external cables and peripherals are disconnected.
- ☐ The necessary tools are available.
- ☐ ESD protection measures are in place.
- ☐ The work area is clean, dry, and well-lit.
- ☐ Screws and components can be stored safely.
- ☐ The PC's user manual or motherboard manual is available if needed.